

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PEE 211 — Past Questions & Materials
by Renance

Course: PEE 211
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 4
Levels: 200

This study pack was compiled and restructured by Renance for PEE 211 students at Federal University of Petroleum Resources, Effurun. It brings together 4 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 4 — PEE 211 INTRODUCTION TO OIL AND GAS INDUSTRY 2024/2025

Level 200 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFUN

COLLEGE OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF PETROLEUM ENGINEERING

2024/2025 First Semester Examination

PEE 211: Introduction to Oil and Gas Industry

For Carry-Over Students Only

Answer ALL Questions. Time Allowed: 1hr. Date: 4/9/2025

100 QUESTIONS

1a). State the main five sectors of the petroleum downstream. Briefly describe the American Petroleum Institute (API). [20] / ENGINEERING

a

1b). Why is the petroleum industry considered to be so-complicated?

2b). List the even main aspects of the oil and natural gas value chain? [10]

2a). How do we determine where to drill and predict what we will find BEFORE we start drilling?

3b). What would one do to maintain a healthy petroleum company? [10]

4b). What is the difference between a "Refinery" and "Oil Refining"? [10]

(4) [10] SETA

5a). Companies in the Petroleum industry can be subdivided into several major categories. Name them. [10]

6b). State in no particular order, the top 5 largest world oil companies by reserves and production. [10]

7b). Differentiate between "Refined Product" and "Petrochemical Product". [10]

8a). What do you understand by the term "The Seven Sisters"? [10]

9b). What do you understand by the term "New Seven Sisters"? [10]

10b). What are Service Petroleum Companies? [10]

11b). Explain the concept of Crude Oil Processing. [10]

12b). Explain the concept of Refining. [10]

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Paper 2 of 4 — PEE 211 OVERVIEW OF OIL AND GAS INDUSTRY 2023,2024 Past Questions.

Level 200 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

_ COLLEGE OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF PETROLEUM ENGINEERING

2023/2024 First Semester Examination

PEE 211: Overview of Oil and Gas Industry

Answer ALL Questions. Time Allowed: thr.30mins. Date: 20/05/2024

QUESTIONS

1. (a) Describe the business cycle of the upstream oil and gas sector, and what are its (as in the upstream sector), What is the downstream sector?

(b) Differentiate between International Oil Companies (IOCs) and National Oil Companies (NOCs), providing examples of each.

(c) Briefly describe the process of distillation of crude oil in petroleum refining and explain the main products obtained from this process.

(d) Discuss the activities of the midstream sector in the oil and gas industry.

2. (a) Explain the role of the oil and gas industry in the Nigerian economy.

(b) Discuss the impact of the oil and gas industry on the environment.

(c) Briefly discuss the structure and functions of the oil and gas industry, and provide examples of midstream activities. (7 marks)

(d) What is the significance of the "Seven Sisters"?

(e) Briefly discuss their historical composition.

3. (a) Discuss the role of the oil and gas industry in the Nigerian economy.

(b) What are the roles of the oil and gas industry? Provide examples of their roles.

(c) Discuss the role of the oil and gas industry in the Nigerian economy.

(d) Discuss the role of the oil and gas industry in the Nigerian economy.

(e) What are the main components of the oil and gas industry, and how are refined products produced?

(f) Discuss the role of the oil and gas industry in the Nigerian economy.

(g) Provide an American Petroleum Institute (API) classification of the petroleum industry and explain its significance.

(h) What is the difference between a "Refinery" and "Oil Refining"?

Aai). Explain the business targets of upstream oil companies and how they align with the overall objectives of the industry.

ii), Describe the different types of exploration and production companies in the oil and gas industry, and discuss their respective roles

" and "Petrochemical Product".

b). Differentiate between "Refined Product

Paper 3 of 4 — PEE 211 PAST QUESTIONS

Level 200 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

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COLLEGE OF ENGINEERING & TECHNOLOGY

2020/2021 First Semester Examination

PEE 211: Introduction to Oil and Gas Industry

Answer Question 1 and any other two Questions. Time Allowed: 1 hour:

QUESTIONS

1a) State the main sectors of the petroleum industry.

1) What is an oil refinery?

1b) What relates drilling engineering and production engineering?

1) Why must casing be perforated? Which casing is considered as the final casing for most wells?

2a) Who is a Drilling Engineer? Explain the two main functions of the drilling engineer.

1) What is referred to as a "Trap"?

1b) What causes blowout during drilling operations? State the effects of blowout.

3a) What is an Exploration wells (or wildcat wells)?

ii) List the natural gas utilization opportunities in Nigeria.

b) What makes natural gas different from other fossil fuels?

4a) What is an integrated company?

i) What is an independent company?

b) What are the physical characteristics of Natural Gas?

COLLEGE OF ENGINEERING & TECHNOLOGY

2019/2020 First Semester Examination

» FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PEE 211: Introduction to Oil and Gas Industry

Time Allowed: 1hr. 30mins Date: 16/04/2021

QUESTIONS

1a) State the main sectors of the petroleum industry.

1) Why is there need for alternate hydrocarbon sources?

a

1b) What is the first period in the producing life of a reservoir?

1) State the three natural forces that move the fluids in a reservoir,

Answer ALL Questions.

2a1) List the six components of a rotary drilling rig, Stating one equipment that each of the components

is made of,

). List the personnel on a rotary drilling rig.

bi). What is the difference between a Servicing company and an Operating company? State two examples each.

i), Differentiate between a Kick and a Blowout.

3a1). What are the 3 ways in which unconventional supplies differ from conventional supplies of oil and

natural gas?

1). State the physical characteristics of natural gas. 2

bi) Why is natural gas considered as one of the cleanest forms of gas? we

i Explain the need for treating crude oil,

4a1) What is Revenue?

1). What is the difference between Direct expenditures and Indirect expenditures?

Table below shows a company's investment of \$10,000.00, which is estimated to generate

5 years period, with an operating expenses of \$14,000.00 per

(FIT) of \$1000. Compute the Net Cash-Flow and the Cumulative

as shown in the table.

b). The table

the \$20,000.00 annually over a

period and a Federal Income Tax

Net Cash Flow over the 5 year period

Investment (\$) | Expenses (\$) | Revenue (\$) FIT (\$)

Year 0 | 10,000.00 |

Year 1 | 14,000.00 | 20,000.00 100000 |

Year 2 | 14,000.00 | 20,000.00 | 1,000.00

Year 3 | 14,000.00 | 20,000.00 | 1,000.00

Year 4 | 14,000.00 | 20,000.00 | 1,000.00

Year 5 | 14,000.00 | 20,000.00 | 1,000.00

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF PETROLEUM ENGINEERING

2022/2023 First Semester Examination

PEE 211: Introduction to Oil and Gas Industry

Time Allowed: 1hr. Date: 27/05/2023

QUESTIONS

Answer ALL Questions.

1ai), State the main five sectors of the petroleum industry as classified by the American Petroleum Institute (API). .

ii), Why is the petroleum industry considered to be so complicated?

b). List the seven main aspects of the oil and natural gas value chain?

2ai). How can we determine where to drill and predict what we will find BEFORE we start drilling?

li). What would one do to maintain a healthy petroleum company?

b). What is the difference between a "Refinery" and "Oil Refining"?

3ai). Companies in the Petroleum industry can be subdivided into four major categories. State them.

li). State in no particular order, the top 5 largest world oil companies by reserves and production.

b). Differentiate between "Refined Product" and "Petrochemical Product".

4ai). What do you understand by the term "The Seven Sisters"?

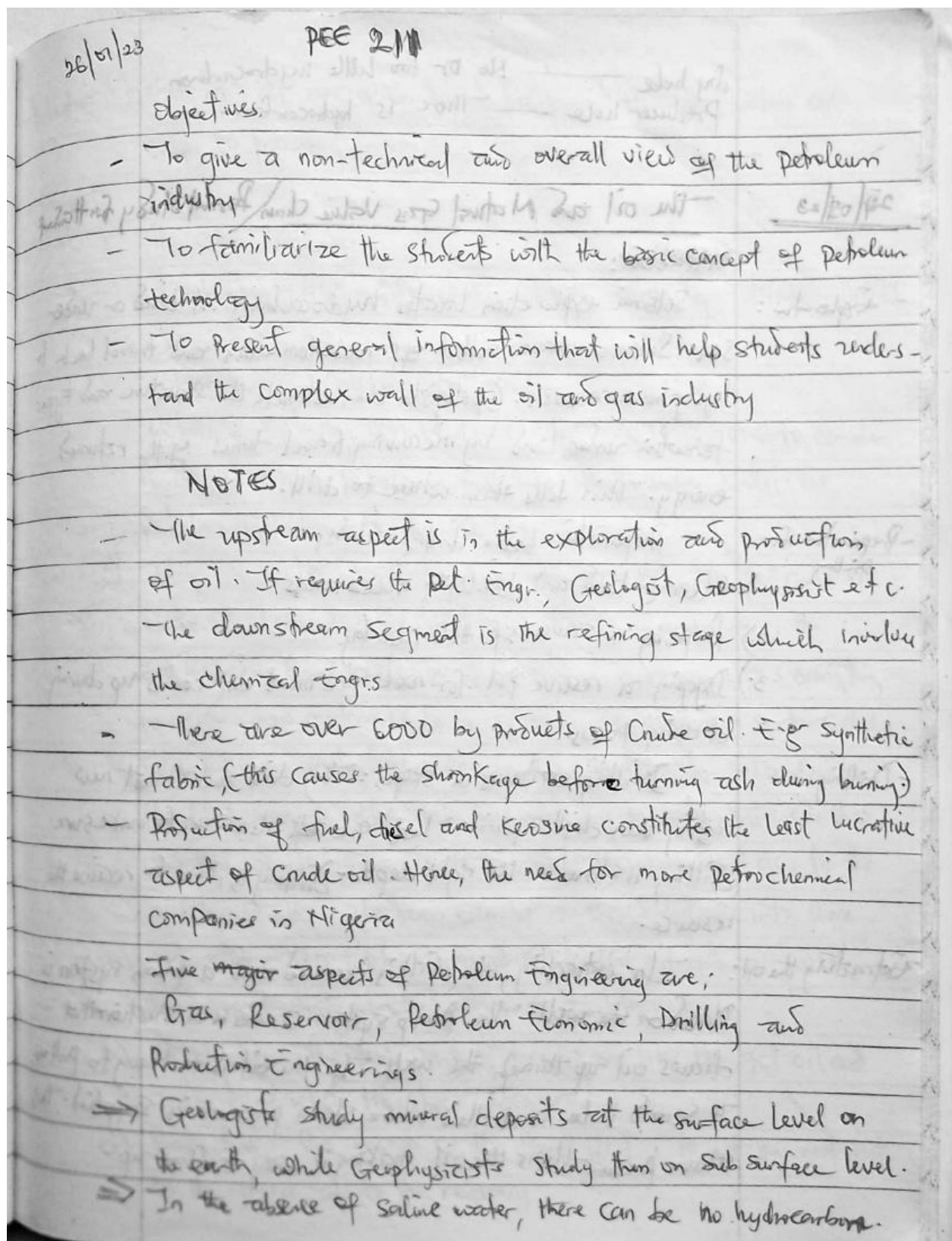
li). What do you understand by the term "New Seven Sisters"?

b). What are Service Petroleum Companies?

Paper 4 of 4 — PEE 211 handout

Level 200 • First Semester • Handout

This source paper is a scanned document; its pages are reproduced below.



dry hole — No or too little hydrocarbon
 Producer hole — There is hydrocarbon

25/05/23

The oil and Natural Gas Value chain ~~Personal study~~ further

~~Exploration~~

- Exploration: Seismic exploration locates hydrocarbons on land or under sea. Seismic waves reflect off rock formations and travel back to hydrophone receivers. Geologists then estimate the structure and types of formation under land by measuring travel times of the reflected energy. This tells them where to drill.

- Preparation for Drilling

Preparation before drilling includes:

- 1.) Clearing land and building access roads
- 2.) Preparing a source of H₂O nearby.
- 3.) Digging a reserve pit for rock and mud that comes up during drilling process.

- Drilling:

Drilling continues in stages: They drill and cement new casing then drill again. Tests are also run to make sure drilling is done on the right depth. Drilling is done to receive the resource.

- Extracting the oil:

In extracting, the drill is removed and a pump system is placed on the well. The pump system creates a suction that draws oil up through the well. If the oil is too heavy to pull up, a second hole is drilled where steam pressure is supplied. This steam pressure thins the oil making it easy to flow up.

- Production: Gas and oil are gathered and transported through pipelines or ships to processing facilities.

- transport
- marketing

The chain includes:

Exploration - Drilling & Production - Shipping - Refining - Storage - Distribution

Petroleum Industry Structure

Petroleum industry can be divided by five ways, one to consider is the American Petroleum Institute which further divides the petroleum industry into five sectors;

- 1) Upstream sector (Exploration, development and production of crude oil and natural gas): The upstream sector is also known as the E&P sector (i.e. exploration & production). Upstream covers everything to getting raw materials to a refinery. The upstream sector includes: Searching for potential underground or underwater oil and gas fields, drilling of wells and subsequently operating the wells that recover and bring the crude oil and/or raw natural gas to the surface. The upstream can be further subdivide into three:
 - a) Exploration: This part is focused on finding oil and gas 'pools'; ^{which} ~~the~~ requires? which blocks? where on the block?
 - b) Development: The 2nd part is focused on how to get oil and gas out of what has been discovered.
 - c) Production: The mission of the third part is to get the most out of the ground and to the refinery.

Upstream oil company target include:

- Replacing production with new reserves
- Keep finding cost low e.g below \$1.

2. Midstream (Transportation, Processing, Storage and Distribution):

3. Downstream (Manufacturing, Refining & Petro-chemicals, wholesales marketing.)

4) Pipeline Sector

5) Service or supply sector

Petroleum Refining:

Crude oil from the field is a mix of hydrocarbons of different molecular length). Refining is the process in which various components of crude oil are separated. Crude oil refining is the process of transforming crude oil into finished petroleum products (that the market demands.)

A refinery is a plant where crude oil is boiled and distilled to separate the individual components.

- Atmospheric distillation is essential process from which refining starts. It is followed by other stages (Vacuum distillation, Cracking: thermal or catalytic, etc.) - the objective is to increase the output of light product which are more valuable, and reduce residuals, which constitute a problem.

International oil companies (IOCs)

Reserve size is not the only way to divide the industry (It is most often used in reference to NOCs). While reserve size and industry segment are ^{both} used to describe ~~IOCs~~ IOCs. The American Petroleum Institute divides the industry into 5 categories ^{based on function} and these divisions help to explain why having large petroleum reserve does not automatically translate into large revenue. The industry segments are: Upstream, Downstream, Pipeline, Marine, Service & Supply (General).

National oil companies (NOCs)

- These are state agencies and are set up much like any IOC
- The major difference is that IOCs ^{release} ~~release~~ earnings reports and have stock holders. Prior to now, IOCs were the major producers, ~~in~~ now NOCs have been organized in most countries with large oil reserve. This trend has occurred for two reasons:
 1. The rise of NOCs in politics: Countries in which large oil reserves can be found have slowly wrested away the rights of IOCs that initially controlled the oil.
 2. The rise of NOCs in industrial progress: Many of the oil-rich nations have leveraged their tremendous natural resources to negotiate profitable contracts with IOCs for extraction and development.
- The creation of OPEC contributed to this, the development of their own means of extracting and refining oil

has also allowed NOCs to reduce their ~~all~~ reliance on IOCs.

Operator Companies (or Exploration and Production (E&P) Companies).

These are companies that decide where and how to drill for hydrocarbons, and own the rights once produced.

E.g: BP, Chevron, Exxon mobile, Shell, Noble energy, Anadarko etc

Types of Exploration and Production Companies.

1. Integrated/Supermajors (Upstream and downstream):

It is one that does everything from finding oil and gas reserves to sales to customers. It breaks up the entire process into two main stages:

- Upstream: covers everything ~~from~~ ^{to} getting raw material to a refinery
- Downstream is everything from refining to sales.

2. Exploration and Production/Independents (just upstream):

3. National oil companies (NOC)

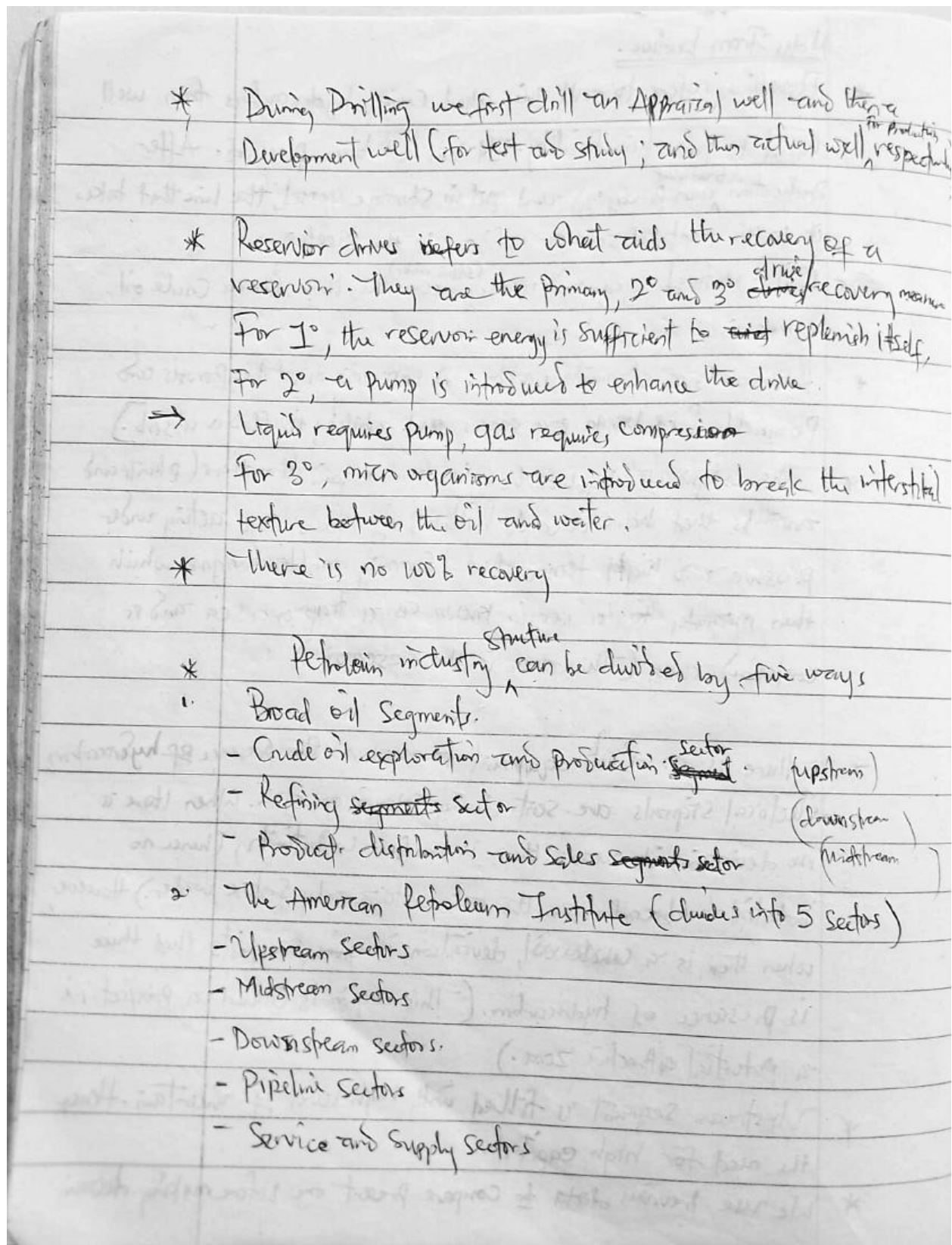
The most influential national oil and gas companies based in countries outside ^{the} OECD (Organization of Economic Cooperation and Development) are labeled "New seven ~~six~~ sisters".

This group comprises:

1. Saudi Aramco (Saudi Arabia) (4) PDVSA (Venezuela) (7) Petromex (Mexico)
2. China National Petroleum Corporation (5) Petrobras (Brazil)
3. Gazprom (Russia) (6) National Iranian oil Company

Notes From Lecture.

- * Flowlines refers to all lines that convey hydrocarbons from well head to production facility. It has not been processed. After production ^(treatment) and processing and put in storage vessel, the line that takes it to the point of sale or refinery is the pipeline.
- * Oil, Natural gas, and water ^(saline water) are the three phases crude oil comes in.
- * Reservoir is a sedimentary rock. A reservoir must be porous and permeable [i.e. large pore spaces and ability to flow or absorb.]
- * The origin of hydrocarbon is from organic matters (plants and animals that have decayed billions of years ago acting under pressure and high temperature forming molten magma which then migrates to a region known as a trap or a sea and is contained area (this area is the reservoir.)
- * There is no direct equipment to measure the presence of hydrocarbon, electrical signals are sent during seismic operation. When there is no deviation, it means there is uniform conductivity (here no potential hydrocarbon as the rock contains only saline water.) However, when there is a ~~considerable~~ deviation, it ~~gives~~ suggests that there is presence of hydrocarbon. (This region is called a prospect i.e. a potential attractive zone.)
- * Upstream segment is filled with high level of uncertain. Hence the need for high capital.
- * We use previous data to compare present one before making decision



3. The oil industry can be divided into four major company types:
- NDCs e.g. NNPC belonging to Nigeria; Saudi Aramco belonging to Saudi Arabia
 - IOCs e.g. Chevron, Exxon mobile, Shell etc. having ^{branch} activities across the world.
 - Operator Companies (E&P company). They run operations for the lifetime of ^{a company} $\wedge$
 - Service Companies. They provide services and move on.

4. Main categories of participants in international oil market.

- NDCs
- IOCs ^{major} and their trading arms.
- Independent oil trading company
- Financial houses and non industry speculators.

5. Oil companies used to be classified by sales

- Super-major Companies (the ~~seven~~ sisters: BP, Chevron, Exxon mobile, ConocoPhillips, Shell, Eni and Total SA)
- Majors Companies
- Independents (or jobbers) Companies

- * ~~Crude oil segment~~: Up. Segments in Oil and Gas Industry (3)
- Upstream - Midstream - Downstream.

* Reasons for pet. refining:

- Crude oil is only useful after refining. Over 6,000 products can be derived from crude oil e.g. from synthetic fragments.

- * ^{oil and gas} Reserve refers to what can be produced using today's technology ^{under today's} ~~and present~~ economical conditions.

